

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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PRN No.: 1012411011

Name: Riddhee Kulkarni

Subject: Python Programming

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Lab Assignment: 17

Aim:

Write a program to add two numbers that are given using command line arguments.

Objective:

To understand how to use command line arguments in Python and perform arithmetic operations using user-provided inputs.

Theory:

Command line arguments are values passed to a program at the time of its execution.

In Python, command line arguments are accessed using the built-in **sys module**.

- `sys.argv` is a list that stores command line arguments.
- `sys.argv[0]` contains the program name.
- `sys.argv[1]` and `sys.argv[2]` contain the first and second user inputs.
- The `float()` function is used to convert string inputs into numeric values.
- Exception handling (`try` and `except`) is used to handle invalid inputs.

Algorithm:

- Start the program.
- Import the `sys` module.
- Check if the number of command line arguments is equal to 3.
- If not, display usage message.
- Else, convert arguments into numeric values.
- Add the two numbers.
- Display the result.
- Stop the program.

Pseudo Code:

START

IMPORT `sys` module

IF length of `sys.argv` is not equal to 3

 PRINT usage message

ELSE

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```
TRY
    num1 = convert sys.argv[1] to float
    num2 = convert sys.argv[2] to float
    result = num1 + num2
    PRINT result
CATCH ValueError
    PRINT invalid input message
END
```

Code:

```
import sys;
if len(sys.argv)!=3:
    print("Usage: python program.py<num1><num2>");
else:
    try:
        num1=float(sys.argv[1]);
        num2=float(sys.argv[2]);
        result=num1+num2;
        print(f"Sum of {num1} and {num2} is: {result}");
    except ValueError:
        print("Please enter valid numeric values.");
```

Output:

```
Sum of 10.0 and 20.0 is: 30.0
```

Conclusion:

Thus, the program to add two numbers using command line arguments was successfully implemented and executed. The program correctly handles valid inputs, invalid inputs, and missing arguments using exception handling.